How car design works
Brief video explaining car designing from the sketching stage to the final working model
http://goo.gl/v3xII

Fun Fact
In 1916, 55 percent of the cars in the world were Model T Fords and this record has never been broken since!

the real test will be when this technology has to be brought into practice on the roads, where variables apart from human-driven vehicles such as pedestrians, obstacles or multiple possibly collision-causing cars come into the picture. Also, the system has to be robust enough to work with precision despite communication delays.

Future work includes incorporating driver reaction time data to determine when the system must actively take control of the car and when a passive warning will work. Another aspect being worked upon in the algorithm is including up to eight vehicles to avoid a situation in which speeding up or slowing down to avoid colliding with one car, may lead to collision with another car.

Vehicle-to-vehicle communication
The next frontier for smarter cars would be actual communication with each other. If cars are able to talk to each other, avoiding accidents would be even easier.

Ford, at an event in San Francisco, demoed its intelligent vehicles using Wi-Fi or dedicated short range communication to talk to each other. This vehicle-to-vehicle (V2V) communication takes place on a secured channel allotted by the Federal Communications Commission (FCC) in the US. This way of communication is different from a radar-based system which only identifies hazards that are in front of the vehicle. The technology used by Ford gives the vehicle a 360-degree detection of dangerous situations. Signals from the car's sensors are emitted 10 times a second to provide a 360-degree view around the car. Present demos are being showed using Ford cars, but the car company is working with other automakers, the government as well as local road authorities in the US to create a common communication standard between vehicles across brands.

The technology uses a driver-vehicle interface which sends audible as well as visual prompts to alert the driver of impending dangers. Ford's intelligent vehicles also get advance blind spot warnings, helping the driver anticipate vehicles approaching quickly from behind that will be entering their blind spot.
Video: Ford Intelligent Vehicle Technology video http://goo.gl/RFlt2

Cars parking themselves
The technology we're going to discuss now was developed keeping in mind the elderly as well as physically handicapped populace in mind. For elderly and wheelchair-bound drivers, it's imperative for their car door to completely open to get in and out of the vehicle. In a parking lot, it may be difficult to always find enough space around your vehicle to open the car door completely.

Arne Suppe, Luis Navarro-Serment and Aaron Steinfield from the Robotics Institute of Carnegie Mellon University have developed the Semi-autonomous virtual valet system which allows users to remotely park their cars using the video camera of their iPod touch or iPhone.

The test was conducted on Navlab 11, a Jeep Wrangler, which was fitted with sensors for short-range and mid-range obstacle detection, custom vehicle measurement systems based around GPS, inertial and odometry sensors, and a small array of on-board computers to record and analyse sensor data.

Getting out of the vehicle, the driver scans the parking lot with his/her iPod touch's camera. The laser range detection technology known as LIDAR (Light Detection And Ranging) - using laser scanners mounted on the vehicle - determine an appropriate location to park as well as a path trajectory. The screen has three buttons: Go, Stop and Resume. Using these three commands on their PMP, the driver can control the vehicle. The video is sent to the iPod via a Wi-Fi link for the experiment, but the researchers say that the data link can be selected as per the user's preference (i.e. 3G, 4G wireless, etc.)

Going by the track record of these three researchers - who also developed automatic parallel parking and docking alongside curbs back in the 90s , which is a reality now - a real life commercial rollout of such vehicles can be expected within the next decade.

iPod as a remote control

Videos demonstrating this technology: http://goo.gl/Ve47O

These are just some of the technologies on the verge of being implemented in vehicles in the near future. When we'll get to see them on Indian roads is a matter of debate. But it's an interesting field to explore and one which has a cascading effect on increasing road safety across the country at least at some level.